

AI MENTAL HEALTH

ANA – 4

Report on “Insight summary for stakeholders”

Dataset Used: Adult Psychiatric Morbidity Survey (APMS) 2014

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Insights

Week 1 Analysis Reports (APMS 2014)

1. Dataset Profiling & Quality Assessment

- The APMS 2014 dataset contains ~1,200 records and 21 columns, covering demographic, lifestyle, and mental health indicators.
- Structure is well-organized and suitable for statistical and predictive analysis.
- Missing Values: Moderate gaps (~20% in some age-group columns) were identified; median-based imputation methods were applied.
- Outliers: Certain extreme values were detected, possibly reflecting genuine high-prevalence cases or collection inconsistencies.
- Tools Used: Python, Pandas, and Jupyter Notebook supported profiling, cleaning, and visualization.
- Overall, the dataset provides a strong foundation for mental health trend analysis and modeling.

2. Demographic Breakdown

Age-Based Patterns

- Mental health disorder prevalence varies significantly across age bands.
- Middle-aged adults (30–60 years): Higher average anxiety and depression scores, likely linked to employment stress and family responsibilities.
- Young adults (16–24 years): High variability; one in three young women meets the threshold for a common mental disorder (CMD).
- Older adults (70+): Generally lower average scores, though isolation and chronic illness remain risk factors.

Gender-Based Patterns

- Women consistently report higher anxiety and depression prevalence compared to men.
- Men's scores appear more stable across age groups.
- Gender differences highlight the need for tailored support strategies.

Combined Age + Gender

- Young women (16–24): Highest-risk cohort, with CMD prevalence at 36.1% compared to 16.3% in men of the same age.

- Middle-aged women: Also show elevated risk, reinforcing the importance of gender-sensitive interventions.

Ethnicity-Based Patterns

- Variations observed across ethnic groups, influenced by cultural, economic, and social determinants.
- Barriers such as stigma, language, and unequal access to care may contribute to disparities.

3. Regional Prevalence Mapping Insights

- Mental health disorder prevalence differs geographically, with up to 6.3 percentage points between highest and lowest regions.
- High-burden regions: South West (19.9%), North West (19.0%), London (18.9%), West Midlands (18.6%).
- Medium-burden regions: North East (16.2%, highest anxiety rate at 7.2%), East Midlands, Yorkshire & Humber.
- Low-burden regions: South East (13.6%) and East of England (14.1%), likely due to higher incomes and better primary care access.
- Disorder Mix:
 - Anxiety (GAD) is the most common nationally (6.0%).
 - Depression varies regionally (2.3%–4.6%).
 - Phobias and OCD show localized peaks (e.g., North West phobia 3.1%, East Midlands OCD 2.7%).

4. Key Clinical Insights

- Young women aged 16–24 represent the single highest-risk demographic.

- Regional disparities highlight the importance of localized service planning.
- Disorder-specific peaks (e.g., anxiety in North East, depression in North West) mean one-size-fits-all screening tools may miss critical patterns.
- Visualization techniques (choropleth maps, grouped bar charts) improve clarity for clinicians and planners.

5. Conclusion

Week 1 analysis of APMS 2014 establishes a strong foundation for mental health analytics. Findings emphasize demographic and regional variation in disorder prevalence, with young women and certain regions (South West, North West, London) requiring urgent attention. These insights will directly inform the AI Mental Health Assistant's screening sensitivity, escalation thresholds, and resource recommendations, ensuring that risk is addressed where it is most acute.